ECN 711
Problem Set #4

Prof. Gustavo Ventura.

Due date: September 28th, 2025.

1. Consider the overlapping generations setup with 2-period lives and a perishable consumption good. Preferences are represented by

$$u(c_t(t), c_t(t+1)) = [c_t(t)][c_t(t+1)]^{\beta}.$$

Assume that the cohort size is constant, and that individuals have the same endowment of the consumption good when young and when old, $\bar{\omega}$. There is a government in this economy that consumes $G(t)$, $t = 1, 2, ...$ This government taxes individuals at proportional rate $\theta(t)$. That is, taxes paid for an individual born at t equal $\theta(t)\ \bar{\omega}$ when young, and $\theta(t+1)\ \bar{\omega}$ when old.

1. Suppose that the sequence of government consumption is $G(t) = G_H$ when t is even and $G(t) = G_L$ when t is odd, with $G_H > G_L$.

   Solve for equilibrium allocations and interest rates for all dates.

2. Show that equilibrium interest rates are higher when $G(t) = G_H$. Explain.

3. Suppose instead that there is heterogeneity individual endowments. Do your conclusions from the previous point still hold? Explain.

2. Consider the following overlapping generations economy. Individuals are identical within a generation, live for two periods and population is constant. Individuals can borrow and lend at the rate $R(t)$.

Consumption goods are produced by competitive firms, which have access to a linear technology that uses only labor as an input, with a rental price given by $w(t)$. Output under this technology is given by $Y_t = A(t)L(t)$, with $A(t+1) = A(t)(1+g)$. The consumption goods produced are perishable, and storage is not available.

Differently from exercise 2, individuals have preferences over consumption, c, and *leisure time*, z (non-labor time). Formally,

$$u(c(t), c(t+1), z(t), z(t+1)) = v(c_t(t), z(t)) + \beta\ v(c(t+1), z(t+1)),$$

where $v(.,.)$ and $v(.,.)$ are strictly increasing in both arguments, twice differentiable and strictly concave.

Note that since the total time endowment is 1, then individuals face the constraint $l(t) + z(t) = 1$ for all t.

1. State the decision problem of a young individual. Obtain the first order conditions that characterize the solution to this problem.

2. Define a competitive equilibrium.

3. Suppose from now on that $v(c, z) = \log(c) + \theta \log(z)$. Furthermore, suppose that there is a government that taxes young individuals' income at the proportional rate $\tau(t) \in (0, 1)$. Tax rates obey $\tau(t) = \tau_L$ if t is odd and $\tau(t) = \tau_H$ if t is even, with $\tau_H > \tau_L$. Tax revenues are used for government consumption.

   Solve for equilibrium interest rates for all t.

4. What are the effects of an increase in τ_H on the equilibrium interest rates? Explain.

3. Consider an overlapping generations economy with a perishable consumption good. Population grows at a constant rate, so that $N(t) = N(0)(1+n)^t$ for $t > 0$. All members of a generation t have identical preferences:

$$u(c^h(t), c^h(t+1)) = \log(c^h(t)) + \beta \log(c^h(t+1))$$

Individuals are heterogenous within a cohort and have endowments $\omega^h(t)$ when young, and $\omega^h(t+1)$ when old. These endowments are such that they are *constant* over time, given growth in the size of cohorts. We assume:

$$\frac{\sum_h^{N(t)} \omega^h(t)}{N(t)} = \Omega_y$$

$$\frac{\sum_h^{N(t)} \omega^h(t+1))}{N(t)} = \Omega_o$$

for all t.

There is government in this economy, that imposes a lump-sum tax τ on the young. Tax collections from this tax are used to provide for a transfer b to all individuals when old.

1. State the decision problem of a young individual.

2. Solve for the equilibrium interest rate in this economy, and argue it is constant over time.

3. What is effect of changes in the population growth rate on the equilibrium interest rate? Evaluate the following statement from the perspective of this model: 'Declines in the population growth rate lead to declines in the interest rate.' Explain.

4. Consider the following overlapping generations economy. Individuals live for two periods and cohort size is constant. Individuals receive an endowment when young and old, which grows over time. There are two types of individuals, $h = 1, 2$. Individuals of type 1 constitute a constant fraction of a cohort, ϕ. Their endowment when young is $z^+A(t)$, and when old $z^-A(t+1)$, with

$z^+ > z^-$. Individuals of type 2 constitute a constant fraction of the population $1 - \phi$, and their endowment when young is $z^- A(t)$, and when old $z^+ A(t+1)$. The variable $A(t)$ grows over time, so that $A(t+1) = A(t)(1+g)$ for all t.

There is a market for consumption loans in each period that pays the (gross) rate of return $R(t)$. Individuals value consumption in each period and discount the future at rate $\beta > 0$. An individual $h = 1, 2$ born at time t maximizes

$$U(c^h(t), c^h(t+1)) = \log(c^h(t)) + \beta \log(c^h(t+1)).$$

1. Formulate the problem of a young individual born at time t.

2. Show that in equilibrium there is a unique, constant over time, interest rate despite the growth of the aggregate endowment.

3. What are the effects of an exogenous increase in the fraction of type-1 individuals on the equilibrium interest rate? Derive formally and explain intuitively.

4. Suppose now there is a government in this economy that creates a public pension system at given date. The system taxes the endowment of the young at constant rate τ to finance common benefits to the old, $b(t)$. What are the effects of the presence of the public pension system on the equilibrium interest rate? Derive formally and explain intuitively.